

Rojae Mighty

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Quantitative Research Profile

Physics undergraduate researcher exploring quantitative finance, machine learning, and data-driven modeling alongside nuclear physics, gluon saturation, geometric scaling, and computational methods for high-energy collisions. His work combines Python, numerical analysis, C++/ROOT, and computational modeling.

Education

Stony Brook University, Stony Brook, NY | **B.S. in Physics**, Expected May 2028 | GPA: 3.7

Relevant Coursework: Computational Methods, Statistical Mechanics, Linear Algebra, Calculus, Electromagnetic Theory

Research and Technical Experience

Brookhaven National Laboratory - Upton, NY

DOE Science Undergraduate Laboratory Internship Researcher | June 2025-August 2025; June 2026-August 2026

- Developed ROOT, Python, and Linux scripts to support electron-ion collision research on 10 million simulated events.
- Built reproducible analysis workflows for studying particle behavior and nuclear effects after high-energy collisions.
- Worked independently within an 8-person research team, translating open-ended physics requirements into validated code and organized outputs.

Center for Frontiers in Nuclear Science (CFNS) - Stony Brook, NY

Undergraduate Researcher, Nuclear and High-Energy Physics | January 2025-Present

- Built Python and scikit-learn pipelines to reconstruct key kinematic quantities from 10 million particle-collision events.
- Used C++, ROOT, Monte Carlo event generators, and numerical simulations to study gluon saturation and geometric scaling at the Electron-Ion Collider.
- Created reusable workflows for repeatable analysis, verification, and sharing of research outputs across tasks.
- Presented research at DIS, APS Global Physics Summit, Yale URC, NSBP, APS DNP, and Stony Brook Physics Colloquium.

Quantitative Projects

Market Microstructure Forecasting

GitHub: <https://github.com/iRojae/market-microstructure-forecasting> | January 2025-Present

- Built a Python ML pipeline with pandas, NumPy, and scikit-learn to forecast short-horizon price movement from limit-order-book data.
- Engineered order-flow imbalance, bid-ask spread, and price-change features for market microstructure modeling.
- Evaluated models with walk-forward validation and transaction-cost-aware backtesting to reduce lookahead bias.

J.P. Morgan Quantitative Research Virtual Experience Program - Forage

January 2026

- Completed 4 quantitative research tasks covering commodity analysis, derivatives pricing, credit-risk modeling, and FICO score grouping.
- Modeled natural gas prices using 48 monthly observations and built a storage-contract pricer for a 1,000,000 MMBtu sample contract valued at \$583,118 net.
- Built probability-of-default models on 10,000 borrower records using 6 financial features; evaluated a 7,500/2,500 train-test split.

Honors and Presentations

- 1st Place for Physical Sciences Presentation, Yale Undergraduate Research Conference, January 2026
- Outstanding Oral Presentation Award, National Society of Black Physicists Conference, January 2025
- Barry Barish Undergraduate Research Travel Award and Edward Guiliano Global Fellowship, Stony Brook University, January 2026

Technical Skills

Programming: Python, C++, C, Unix/Linux shell scripting

Data and ML: pandas, NumPy, scikit-learn, feature engineering, statistical analysis, walk-forward validation, dynamic programming

Research Tools: ROOT, Monte Carlo event generators, numerical simulation, Git, GitHub, Linux/Unix environments

Domains: Market microstructure, Electron-Ion Collider physics, particle-collision data, gluon saturation, geometric scaling